

AMAN PARGANIHA

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Experience

Janyu Tech

Mumbai, Maharashtra

Robotics Programming Intern

Aug 2026 – Present

- Own backend delivery on a two-person team building **MyFactory OS**, a Manufacturing Execution System (MES) for industrial-robotics production: an internal shop-floor job tracker plus a customer order-tracking portal.
- Reviewed 12+ MES products (Siemens Opcenter, SAP ME, Tulip, Zetwerk) and wrote the technical scope; engineering leadership approved the build-vs-buy case and minimum viable product (MVP) boundary.
- Designed the relational schema and versioned REST API contract on **FastAPI**, **SQLAlchemy 2.0**, **PostgreSQL 16**, **Alembic**: JWT role-based access, query-level tenant scoping, transactional stage advances behind QC gates, and an append-only audit log.
- Packaged the API, PostgreSQL, MinIO and React/Vite frontend with **Docker Compose**; hosts and credentials come from the environment for on-premise deployment.

Education

International Institute of Information Technology Naya Raipur

India

M.Tech in Computer Science & Engineering (AI/ML) | CGPA: 8.0/10.0

Aug 2025 – Aug 2027

Shri Shankaracharya Technical Campus Bhilai

India

B.Tech in Computer Science & Engineering | CGPA: 8.22/10.0

Aug 2019 – Aug 2023

Technical Skills

Languages: Python, C++, TypeScript, SQL

ML & Deep Learning: PyTorch, Scikit-learn, Transformers, diffusion models, vision-language models, NLP, model evaluation & benchmarking

LLMs & Generative AI: Retrieval-Augmented Generation (RAG) pipelines, LlamaIndex, LangChain, OpenAI API (structured outputs, streaming), prompt engineering, embeddings, vector databases (Qdrant), agentic workflows

Backend & MLOps: FastAPI, Node/Express, REST API design, server-sent events, SQLAlchemy, Alembic, Drizzle ORM, PostgreSQL, JWT auth, Docker Compose, AWS (EC2), MinIO/S3, Inngest, Git/GitHub, GitHub Actions CI

Data & Practices: Pandas, NumPy, ETL pipelines, dataset curation, feature engineering, SQL/NoSQL, unit & end-to-end testing (Vitest, Playwright), code review, system design

Projects

SayShip — Prompt-to-App Generation Platform | [Code](#) | [Live Demo](#)

TypeScript, React 19, Express 5, PostgreSQL, Drizzle, esbuild, OpenAI

- Shipped a platform that turns a plain-English prompt into a running full-stack app: a planner agent writes a structured plan with **OpenAI Structured Outputs**, a writer agent streams files over **SSE**, an esbuild check blocks broken builds, and a repair pass returns compiler `file:line` errors to the model.
- Ran untrusted generated code in an **opaque-origin sandbox** (iframe without `allow-same-origin`, plus a CSP `sandbox` header) with a capability-keyed CRUD API per app, isolating generated apps from platform sessions and data; added immutable versioning and an atomic DB lease for one run per project.
- Covered the production build with **67 Vitest/supertest** and **10 Playwright E2E tests** in GitHub Actions CI; a mock LLM provider lets the product and the full test suite run deterministically with no API key.

Event-Driven RAG Agent for Document Ingestion & Q&A | [Code](#) | [Live Demo](#)

Python, FastAPI, Inngest, LlamaIndex, Qdrant, OpenAI, Docker

- Built an event-driven RAG system (FastAPI backend, Streamlit chat UI, Qdrant in Docker) that ingests multi-page PDFs asynchronously and answers questions with grounded, cited responses.
- Made ingestion durable and fault-tolerant with Inngest, using `parse` → `embed` → `store` steps that retry independently, over LlamaIndex chunking, OpenAI embeddings and a Qdrant store; chunking and embedding strategies stay swappable.

Multimodal Credit Risk Analysis using SEC XBRL & NLP | [Code](#) | [Live Demo](#)

Python, Pandas, Scikit-learn, NLTK/VADER

- Wrote an end-to-end pipeline turning ~41M raw SEC XBRL records (2022 – 2024) into a curated dataset of ~35,000 companies with 47 features (34 financial, 13 from NLP).
- Compared Random Forest, Gradient Boosting, Logistic Regression and SVM, reaching **97.9% binary** / **93.8% multi-class accuracy** on financial features; per-feature-set evaluation and leakage checks exposed and discarded an inflated multimodal result.

Research

LiveJEPA: Real-Time Vision-Language Captioning (M.Tech thesis, in progress | PyTorch) — reconstructing Meta FAIR’s VL-JEPA pipeline for streaming video captioning: frozen V-JEPA 2 encoder with a lightweight predictor head aligned to the SONAR text-embedding space, evaluated on EgoExo4D.

DDPM from Scratch: Denoising Diffusion Probabilistic Model ([Code](#) | PyTorch) — reimplemented DDPM from the paper without diffusion libraries: noise schedules, closed-form forward diffusion, UNet denoiser with sinusoidal timestep embeddings, and ϵ -prediction training with ancestral sampling, trained on Fashion-MNIST.

GreenCloudRL: Meta-RL for Energy-Efficient Cloud Scheduling (manuscript submitted) — hierarchical meta-RL scheduler (A2C/PPO with Reptile) with SHAP-based explainability, achieving ~14% energy reduction on Google and Alibaba cluster traces.

Open Source Contributions

Matplotlib — PR #31422 merged and shipped in v3.11.0: resolved issue #26620 by improving legend parameter documentation, reviewed and approved by core maintainer timhoffm.

JupyterLab — PR #18610 merged, PR #18668 merged: improved user-facing documentation and added Playwright end-to-end test coverage, both cleared multi-round maintainer review.

Certifications & Achievements

GATE CSE: Top 7.8% of ~210,000 candidates nationwide — secured M.Tech (AI/ML) at IIIT Naya Raipur. | **Problem Solving:** 200+ LeetCode problems solved. | **Google × Kaggle** — **5-Day AI Agents Intensive:** agentic LLM workflows, LlamaIndex/LangChain orchestration, RAG system design.